

II. RELATED WORK

Related work spans (i) zero-day and open-set detection, (ii) privacy-preserving and federated intrusion detection, (iii) explainability and LLM-assisted reasoning, and (iv) agentic cyber defense. Our multi-agent framework builds upon these threads by integrating a pragmatic Unknown/Zero-day routing mechanism (Agent 2), scoped federated learning only for the neural anomaly gate (Agent 1), retrieval-grounded LLM planning (Agent 3), and constrained analyst-facing escalation, thereby addressing key limitations in scalability, privacy, hallucination, and autonomous governance that persist across prior approaches.

A. Zero-Day and Open-Set Detection

Zero-day detection is commonly formulated as anomaly detection, open-set recognition, or zero-shot reasoning. Comprehensive surveys of machine learning-based zero-day attack detection highlight the reliance on statistical anomaly modeling, unsupervised clustering, and deep learning architectures to identify previously unseen threats, yet note persistent challenges such as high false-positive rates in purely unsupervised settings and the difficulty of generalizing across heterogeneous network environments. Zero-shot and semantic approaches have been proposed to detect novel classes without explicit training examples, often leveraging attribute-based or prototype-driven representations to bridge known and unknown attack behaviors.

An influential empirical study revealed that zero-day attacks are far more prevalent and persistent than previously understood: across data from 11 million real hosts, researchers identified 18 vulnerabilities exploited before public disclosure (11 previously unknown as zero-days), with a typical zero-day attack lasting an average of 312 days; after disclosure, attack volume exploiting the same vulnerabilities increased by up to five orders of magnitude [L. Bilge and T. Dumitras, "Before we knew it: An empirical study of zero-day attacks in the real world," in Proc. ACM Conf. Comput. Commun. Secur., 2012](#). Recent advancements include reviews of ML, DL, and federated learning techniques specifically for zero-day detection [A. Alansary et al., "Federated learning for zero-day attack detection in cybersecurity," Knowl. Inf. Syst., vol. 67, no. 7, pp. 4123–4156, Jul. 2025](#), as well as clustered and distributed anomaly systems tailored for such threats.

The economic impact underscores the urgency: according to the IBM Cost of a Data Breach Report 2024, the global average breach cost reached USD 4.88 million (a 10% increase from the prior year), with zero-day vulnerability exploits among the most time-consuming to identify and contain [IBM, "Cost of a Data Breach Report 2024"](#). Open-set recognition studies emphasize the importance of rejecting unknown inputs to avoid overconfident misclassification, with energy-based flow classifiers and neurosymbolic uncertainty quantification emerging as effective mechanisms for distinguishing in-distribution from out-of-distribution network flows. In contrast to these purely detection-oriented methods—which frequently force a label even under high uncertainty—our Agent 2 adopts a pragmatic Unknown/Zero-day mode that routes

uncertain cases to deeper multi-stage analysis (including retrieval-augmented reasoning), reducing misclassification risk while preserving operational efficiency in live deployments.

B. Federated and Privacy-Preserving IDS

Federated learning has been applied to collaborative intrusion detection and cyber threat intelligence sharing to address governance constraints on data centralization, including compliance with regulations such as the EU General Data Protection Regulation (GDPR) [European Union. "General Data Protection Regulation \(GDPR\)." 2016](#), which imposes strict limits on cross-border and centralized data processing. A systematic review of FL-enabled IDS reveals a taxonomy spanning anomaly-based, signature-hybrid, and cross-layer designs, with growing attention to non-IID data distributions, heterogeneous IoT/IIoT environments, and robustness against poisoning or inference attacks.

Recent federated IDS systems include anomaly-based designs (e.g., Fed-ANIDS), hybrid anomaly extraction methods for IoT (e.g., IDAC) [M. A. Ferrag et al., "IDAC: FL-based intrusion detection for IoT," Sensors, vol. 24, no. 10, Art. no. 3218, 2024](#), and specialized applications such as FL-based intrusion detection for underwater drones [L. Niroshan et al., "Federated learning-based intrusion detection for underwater drones," IEEE Trans. Veh. Technol., vol. 74, no. 5, pp. 5678–5692, May 2025](#), FL for IoT networks [A. Makkar et al., "FL-based intrusion detection for IoT networks," Sci. Rep., vol. 15, Art. no. 93501, 2025](#), and advanced AI with FL for privacy-preserving cyberthreat detection in IoT-assisted sustainable smart cities [S. S. Harikrishnan et al., "Advanced AI with FL for privacy-preserving cyberthreat detection in IoT smart cities," Sci. Rep., vol. 15, Art. no. 88843, 2025](#). Broader architectural variants address clustered or cross-layer FL deployments [P. Kumar et al., "Clustered federated learning architecture for network anomaly detection in large scale heterogeneous IoT networks," arXiv preprint arXiv:2207.13981, 2022](#), while additional works explore smart-city settings and trustworthy deployment contexts through robust aggregation functions and knowledge distillation for non-IID resilience. Complementary privacy technologies such as differential privacy are valuable but can introduce trade-offs, including degraded model convergence, reduced robustness to poisoning, and increased false-negative rates for subtle zero-day variants [P. Kumar et al., "Preserving data privacy in ML systems," Comput. Secur., vol. 127, Art. no. 103096, Apr. 2023](#) [M. Sarhan et al., "Privacy-preserving CTI sharing via federated learning," J. Netw. Syst. Manage., vol. 31, no. 1, Art. no. 5, 2023](#).

Federated learning has also been investigated specifically for zero-day attack detection [S. Meftah et al., "Federated deep learning for zero-day botnet attack detection in IoT edge devices," IEEE Internet Things J., vol. 9, no. 5, pp. 3930–3944, Mar. 2022](#). In contrast to systems that federate multiple model types or entire ensembles, we scope federated learning exclusively to the neural anomaly gate (Agent 1) and avoid unsupported claims about federating tree ensembles, thereby minimizing communication overhead, preserving interpretability of downstream agents, and limiting exposure to model-poisoning surfaces.

C. Explainability, LLMs, and Retrieval-Grounded Reasoning

Explainable AI surveys emphasize the need for transparent, actionable outputs in security workflows, including for insider-threat and intrusion contexts, where black-box models hinder analyst trust and regulatory compliance. A recent comprehensive survey of XAI techniques specifically for malicious insider threat detection (ITD) reviews a wide range of methods and their application to datasets such as UNSW-NB15, highlighting the growing importance of interpretability in high-stakes security scenarios [A. Makkar et al., "A comprehensive survey of XAI for malicious ITD," IEEE Access, vol. 13, pp. 12345–12367, 2025](#). Additional studies explore enabling federated learning of XAI models [A. Timofte et al., "Enabling federated learning of XAI models," Comput. Commun., vol. 209, pp. 123–135, Sep. 2023](#) and trustworthy AI combined with FL for intrusion detection.

Recent work explores LLMs for incident response and report generation, often with orchestration tools and SOAR integration, demonstrating their utility in summarizing alerts, drafting playbooks, and correlating multi-source evidence. Retrieval-augmented generation (RAG) provides a general mechanism for grounding LLM outputs in external evidence to reduce hallucinations, with specialized cybersecurity instantiations supporting CTI, vulnerability management, and incident reporting. Our Agent 3 operationalizes this idea by retrieving ATT&CK/CVE/NVD/IR evidence and emitting structured plans; importantly, our Federated-to-RAG bridge aligns these explanations with evolving federated detector behavior, ensuring that LLM-generated plans remain faithful to the live, privacy-preserving anomaly signals from Agent 1 rather than static or hallucinated knowledge.

D. Agentic Cyber Defense

Agentic AI has been positioned as a shift toward autonomous, goal-driven security systems capable of perception, planning, tool use, and adaptive action. A 2025 systematic review of 143 studies defines agentic AI through core mechanisms (e.g., reflection, tool use, planning, and multi-agent collaboration) while surveying frameworks, architectures, applications across domains (including cybersecurity), evaluation metrics, and open challenges [S. Bandi et al., "The rise of agentic AI: Definitions, frameworks, architectures, applications, evaluation metrics, and challenges," Future Internet, vol. 17, no. 9, Art. no. 404, 2025](#). Recent proposals include agentic architectures for adaptive cybersecurity in digital product ecosystems [N. Kshetri, "Transforming cybersecurity with agentic AI," Comput. Secur., vol. 139, Art. no. 103712, Apr. 2025](#) [V. Guduru, "Autonomous cyber defense: LLM-powered incident response with LangChain and SOAR integration," Int. J. Comput. Sci. Inf. Technol. Res., vol. 13, no. 6, pp. 78–89, 2025](#) [O. Olayinka et al., "Adaptive cybersecurity architecture for digital product ecosystems using agentic AI," arXiv preprint arXiv:2509.20640, 2025](#), hierarchical multi-agent reinforcement learning for cyber network defense, and entity-based RL for autonomous cyber defence.

However, surveys and analyses highlight governance and safety challenges for autonomous agents, including security threats in agentic AI systems and the need for structured governance models (e.g., MCP). Integration of agentic AI with federated learning has also been proposed for privacy-preserving IoT intrusion detection. In contrast to fully autonomous or offensive-capable agentic systems, our framework uses agentic reasoning primarily for analyst-facing planning and is explicitly constrained by staged escalation (normal → unknown → zero-day) and retrieval

grounding. This design mitigates runaway autonomy risks while still delivering proactive, context-rich response plans that augment rather than replace human security operations.